

# Exercise 2

Table of task 1

|         | Meyda audio features | Justification                                                                                                                                                                                                                                                                              |
|---------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sound 1 | Energy               | There is a lot of noticeable energy changes with the parts of almost silence, changes reflecting this double sound wave in the background beat sound, that value shows those changes                                                                                                       |
|         | Spectral Centroid    | This feature is great to notice the moments of when that electric piano like sound is clicked, this is more of the front sound we hear periodically – melody                                                                                                                               |
|         | Spectral Slope       | Allows as to see background changes in a breathing change in less significant amount, and more significant changes when electrical piano starts playing, this could be used to draw background colors for more fluctuating changes                                                         |
| Sound 2 | Zcr                  | Really well reflects how the sound pitch is building up in linear way, could smoothly go with slowly growing graphics matching growth of pitch                                                                                                                                             |
|         | energy               | Shows well the changes in loudness of the signal, especially those spikes of almost turbulence hitting sound                                                                                                                                                                               |
|         | Spectral centroid    | Reflects changes in brightness of the sound, from low spectral in the beginning going with active spikes of values throughout to higher spectral centroid                                                                                                                                  |
| Sound 3 | energy               | This sound starts with no energy and slowly builds up, so its good to measure that so that could reflect the amount of graphical animation, its not great to have a lot of going on when sound is quiet and does not have much energy at that moment, this features allows to reflect that |
|         | Zcr                  | Allows to show those spinning pitch changes in the audio file                                                                                                                                                                                                                              |

|  |                 |                                                                                          |
|--|-----------------|------------------------------------------------------------------------------------------|
|  | Spectral Spread | Allows to reflect that background slow change between pitch and noise in smoother manner |
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## Description and specification of audio features and mapping implemented in task 2

RMS – it returns values that represent the root mean of the waveform, In my visualization I used it to show when signal is getting loud and there is more action happening in the audio. For mapping I multiplied the value by 850 to get more integer values in range from 0 to couple hundreds. That is how the 4 blue squares that go around the middle are drawn, and nicely reflect heavy beat in later part of music – not showing in the beginning.

Spectral Kurtosis – shows how pointy the spectrum is, in my visualization I used it for red rectangles that change their size and shape to that value. We can see the different between different parts of song when the sound is more pitchy without the beat in the background, making visualization more interesting with different effects through different part of audio track. The way I mapped it is using original values for height but multiplying it by 7 on width to make interesting visual of # pattern.

ZCR – this feature really allows to show those percussive sounds where the main hypnotizing rotating white rectangles grows bigger with every percussive hit and stays calmer thought out more pitched sound that changes is smoother. To make the middle part visible and take over the screen as center point, I multiplied the result value by 6 for mapping.

Link to application:

<https://hub.labs.coursera.org/connect/sharedztlxrvjl?forceRefresh=false&path=%2FvMBxTlyvUcM5DXb37Upsye6EhaJPW4z7oTLHiHdIrBJcLsAlkbpg8LFmKA6qdMB0%2F&isLabVersioning=true>